Package 'bfpwr'

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Title Power and Sample Size Calculations for Bayes Factor Analysis

Description Implements z-test, t-test, normal moment prior, and binomial test Bayes factors based on summary statistics, along with functionality to perform corresponding power and sample size calculations for fixed and sequential designs as described in Pawel and Held (2025) <doi:10.1080/00031305.2025.2467919>, Pawel and Held (2026) <doi:10.48550/ARXIV.2
ter and Pawel (2026) <doi:10.1177/09622802261507279>.

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`bf01` *Point null z-test Bayes factor*

Description

This function computes the Bayes factor that quantifies the evidence that the data (in the form of an asymptotically normally distributed parameter estimate with standard error) provide for a point null hypothesis with a normal prior assigned to the parameter under the alternative. The standard error is assumed to be known.

Usage

```
bf01(
  estimate,
  se,
  null = 0,
  pm,
  psd,
  log = FALSE,
```

```
  alternative = c("two.sided", "less", "greater")
)
```

Arguments

estimate	Parameter estimate
se	Standard error of the parameter estimate
null	Parameter value under the point null hypothesis. Defaults to 0
pm	Mean of the normal prior assigned to the parameter under the alternative
psd	Standard deviation of the normal prior assigned to the parameter under the alternative. Set to 0 to obtain a point prior at the prior mean
log	Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE
alternative	Direction of the alternative hypothesis, one of "two.sided" (default), "less", or "greater". For a one-sided alternative, the normal analysis prior is restricted to values below null ("less") or above null ("greater") and renormalized. pm and psd describe the normal distribution before truncation. A point prior (psd = 0) must lie strictly on the specified side of null.

Value

Bayes factor in favor of the null hypothesis over the alternative ($BF_{01} > 1$ indicates evidence for the null hypothesis, whereas $BF_{01} < 1$ indicates evidence for the alternative)

Author(s)

Samuel Pawel

Examples

```
bf01(estimate = 0.2, se = 0.05, null = 0, pm = 0, psd = 2)
bf01(estimate = 0.2, se = 0.05, pm = 0, psd = 2, alternative = "greater")
```

---

bfpwrOptions *Numerical settings for Bayes factor designs*

Description

Inspect or change the numerical settings used in subsequent calculations. Explicit function arguments override these session settings.

Usage

```
bfpwrOptions(...)
```

Arguments

...	Named settings, or a single named list of settings. With no arguments, return the effective settings without changing them.

Details

The settings are `ngrid` (sequential integration grid, default `10000`), `tol` (sample-size and t-boundary root tolerance, `1e-8`), `rel.tol` (t-Bayes-factor integration tolerance, `1e-8`), `subdivisions` (integration limit, `1000`), `tail.eps` (omitted t-predictive tail mass, `1e-6`), and `tail.nquad` (t-tail quadrature nodes per dimension, `512`). The absolute integration tolerance follows `rel.tol` unless `abs.tol` is supplied to a calculation. The internal z-boundary tolerance is fixed separately to preserve accuracy near point priors.

Values are stored in R options named `bfpwr.ngrid`, `bfpwr.tol`, and so on. Changes last for the current R session. Saved design objects retain the controls needed to reproduce their plots.

Value

A named list of effective settings. When setting values, the previous settings are returned invisibly, so they can be restored with another call.

Examples

```
bfpwrOptions()
old <- bfpwrOptions(ngrid = 20000, rel.tol = 1e-9)
bfpwrOptions(old)
```

`binbf01` *Binomial Bayes factor*

Description

This function computes the Bayes factor for testing a binomial proportion p based on x observed successes out of n trials. Two types of tests are available:

- Test of a point null hypothesis: The Bayes factor quantifies the evidence for $H_0 \colon p = p_0$ against $H_1 \colon p \neq p_0$. A beta prior is assigned to the proportion p under the alternative hypothesis H_1.
- Test of a directional null hypothesis: The Bayes factor quantifies the evidence for $H_0 \colon p \leq p_0$ against $H_1 \colon p > p_0$. A beta prior that is truncated to the range $[0, p_0]$ under the null H_0 and to $(p_0, 1]$ under the alternative H_1 is assigned to the proportion p under the corresponding hypothesis.

Usage

```
binbf01(
  x,
  n,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  log = FALSE
)
```

Arguments

x	Number of successes
n	Number of trials
p0	Tested binomial proportion. Defaults to 0.5
type	Type of test. Can be "point" or "direction". Defaults to "point"
a	Number of successes parameter of the beta prior distribution. Defaults to 1
b	Number of failures parameter of the beta prior distribution. Defaults to 1
log	Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to FALSE

Value

Bayes factor in favor of the null hypothesis over the alternative ($BF_{01} > 1$ indicates evidence for the null hypothesis, whereas $BF_{01} < 1$ indicates evidence for the alternative)

Author(s)

Samuel Pawel

See Also

pbinbf01, nbinbf01

Examples

```
## example on Mendelian inheritance from ?stats::binom.test
binbf01(x = 682, n = 925, p0 = 3/4, a = 1, b = 1, type = "point")
## 18.6 => strong evidence for the hypothesized p = 3/4 compared to other p

## with directional hypothesis
binbf01(x = 682, n = 925, p0 = 3/4, a = 1, b = 1, type = "direction")
## 1.5 => only anecdotal evidence for p <= 3/4 over p > 3/4

## Particle-counting experiment from Stone (1997) with point null
binbf01(x = 106298, n = 527135, p0 = 0.2, a = 1, b = 1, type = "point")
## 8.1 => moderate evidence for the alternative over the null
```

```
## Coin flip experiment from Bartos et al. (2023) with point null
binbf01(x = 178079, n = 350757 , p0 = 0.5, a = 5100, b = 4900, type = "point")
## => 1/1.72e+17 extreme evidence in favor of the alternative over the null
```

`dirbf01` *Directional z-test Bayes factor*

Description

This function computes the Bayes factor that quantifies the evidence that the data (in the form of an asymptotically normally distributed parameter estimate with standard error) provide for a directional null hypothesis that the parameter value is less than or equal to the null value against the alternative that it is greater than the null value. A marginal normal prior is assigned to the parameter and split at the null value into the corresponding lower-side null and upper-side alternative distributions. The standard error is assumed to be known.

Usage

```
dirbf01(estimate, se, null = 0, pm, psd, log = FALSE)
```

Arguments

`estimate`	Parameter estimate
`se`	Standard error of the parameter estimate
`null`	Null value that separates the null from the alternative hypothesis. Defaults to `0`
`pm`	Mean of the normal prior assigned to the parameter
`psd`	Standard deviation of the normal prior assigned to the parameter
`log`	Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to `FALSE`

Value

Bayes factor in favor of the null hypothesis over the alternative ($BF_{01} > 1$ indicates evidence for the null hypothesis, whereas $BF_{01} < 1$ indicates evidence for the alternative)

Author(s)

Samuel Pawel

Examples

```
dirbf01(estimate = 0.2, se = 0.2, null = 0, pm = 0, psd = 2)
```

nbf01 *Sample size determination for z-test Bayes factor*

Description

This function computes the required sample size to obtain a Bayes factor (bf01) more extreme than a threshold k with a specified target power.

Usage

```
nbf01(
  k,
  power,
  usd,
  null = 0,
  pm,
  psd,
  dpm = pm,
  dpsd = psd,
  nrange = c(1, 10^5),
  lower.tail = TRUE,
  integer = TRUE,
  analytical = TRUE,
  alternative = c("two.sided", "less", "greater"),
  ...
)
```

Arguments

k	Bayes factor threshold
power	Target power
usd	Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$, see details
null	Parameter value under the point null hypothesis. Defaults to 0
pm	Mean of the normal prior assigned to the parameter under the alternative in the analysis
psd	Standard deviation of the normal prior assigned to the parameter under the alternative in the analysis. Set to 0 to obtain a point prior at the prior mean
dpm	Mean of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior pm
dpsd	Standard deviation of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior psd
nrange	Sample size search range over which numerical search is performed. Defaults to c(1, 10^5)

`lower.tail`	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq \mathsf{k})$ (`TRUE`) or $\Pr(\mathrm{BF}_{01} > \mathsf{k})$ (`FALSE`) should be computed. Defaults to `TRUE`
`integer`	Logical indicating whether only integer valued sample sizes should be returned. If `TRUE` the required sample size is rounded to the next larger integer. Defaults to `TRUE`
`analytical`	Logical indicating whether analytical (if available) or numerical method should be used. Defaults to `TRUE`
`alternative`	Direction of the alternative hypothesis, one of `"two.sided"` (default), `"less"`, or `"greater"`. For a one-sided alternative, the normal analysis prior is restricted to values below `null` (`"less"`) or above `null` (`"greater"`) and renormalized. `pm` and `psd` describe the normal distribution before truncation. A point prior (`psd = 0`) must lie strictly on the specified side of `null`.
`...`	Other arguments passed to `stats::uniroot`. The sample-size root tolerance defaults to `tol = 1e-8` and can be overridden.

Details

It is assumed that the standard error of the future parameter estimate is of the form $\mathsf{se} = \mathsf{usd}/\sqrt{\mathsf{n}}$. For example, for normally distributed data with known standard deviation `sd` and two equally sized groups of size `n`, the standard error of an estimated standardized mean difference is $\mathsf{se} = \mathsf{sd}\sqrt{2/n}$, so the corresponding unit standard deviation is $\mathsf{usd} = \mathsf{sd}\sqrt{2}$. See the vignette for more information.

Value

The required sample size to achieve the specified power

Note

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors. The normal design prior remains untruncated for one-sided alternatives. Power need not increase with sample size; for example, the probability of misleading evidence first rises and then falls. If both endpoints of `nrange` are below the target, the numerical search checks the interior maximum. The returned sample size is the smallest at which power reaches the target; power may fall below the target again at larger sample sizes.

Author(s)

Samuel Pawel

See Also

pbf01, powerbf01, bf01

Examples

```
## point alternative (analytical and numerical solution available)
nbf01(k = 1/10, power = 0.9, usd = 1, null = 0, pm = 0.5, psd = 0,
      analytical = c(TRUE, FALSE), integer = FALSE)
```

```
## standardized mean difference (usd = sqrt(2), effective sample size = per group size)
nbf01(k = 1/10, power = 0.9, usd = sqrt(2), null = 0, pm = 0, psd = 1)
## this is the sample size per group (assuming equally sized groups)

## z-transformed correlation (usd = 1, effective sample size = n - 3)
nbf01(k = 1/10, power = 0.9, usd = 1, null = 0, pm = 0.2, psd = 0.5)
## have to add 3 to obtain the actual sample size

## log hazard/odds ratio (usd = 2, effective sample size = total number of events)
nbf01(k = 1/10, power = 0.9, usd = 2, null = 0, pm = 0, psd = sqrt(0.5))
## have to convert the number of events to a sample size
```

---

nbf01seq *Maximum Sample Size Determination for Sequential z-Test Bayes Factors*

Description

Computes the maximum sample size required for a sequential z-test Bayes factor design to reach a target probability of stopping for H_1 or H_0.

Usage

```
nbf01seq(
  k1,
  k0 = 1/k1,
  power,
  usd = sqrt(2),
  null = 0,
  pm = NULL,
  psd,
  dpm = pm,
  dpsd = psd,
  type = c("normal", "directional", "moment"),
  target = c("H1", "H0"),
  nrange = c(2, 10^5),
  looks = 1,
  timing = NULL,
  minN = NULL,
  by = NULL,
  strict = TRUE,
  integer = TRUE,
  search = c("adaptive", "exhaustive"),
  details = FALSE,
  alternative = c("two.sided", "less", "greater"),
  ...
)
```

Arguments

k1	Bayes factor threshold in favor of H_1. Evidence for H_1 is obtained when $\mathrm{BF}_{01} \leq k1$.
k0	Bayes factor threshold in favor of H_0. Evidence for H_0 is obtained when $\mathrm{BF}_{01} \geq k0$.
power	Target stopping probability.
usd	Unit standard deviation, the standard error of the parameter estimate at $n = 1$.
null	Parameter value under the point null hypothesis. Defaults to 0
pm	Analysis prior mean. Not taken into account for type = "moment" for the analysis prior. If dpm is omitted, its default remains pm, including for moment priors
psd	Analysis prior standard deviation (type = "normal" and type = "directional") or scale (type = "moment")
dpm	Mean of the normal design prior
dpsd	Standard deviation of the normal design prior. Set dpsd = 0 to obtain a point prior at dpm
type	Character string. One of • "normal" (default): point null vs. normal alternative (set psd = 0 to obtain a point alternative) • "directional": directional null $\theta \leq$ null vs. directional alternative $\theta >$ null with a marginal normal prior • "moment": point null vs. normal moment alternative which is centered around null One-sided alternative values are supported for type = "normal" and type = "moment"; type = "directional" compares two composite hypotheses and retains its existing meaning.
target	Character string. Either "H1" for the final cumulative probability of stopping for H_1, or "H0" for the final cumulative probability of stopping for H_0.
nrange	Maximum sample size search range over which numerical search is performed. Defaults to c(2, 10^5).
looks	Number of sequential looks when timing is not supplied. The default looks = 1 gives a fixed design.
timing	Optional cumulative information fractions for the looks. Values must be positive, strictly increasing, no larger than one, and end at one. Candidate look sample sizes are ceiling(maximumN * timing - sqrt(.Machine$double.eps)) with the final look fixed to maximumN.
minN	First-look sample size when by is supplied. Defaults to the lower search bound.
by	Optional sample-size increase between looks. If supplied, schedules are generated as seq(minN, maximumN, by = by) with maximumN appended as the final look.
strict	Logical. If TRUE and there are more than two critical values per stage, enumerate all possible region combinations. If FALSE, only integrate over the main regions where the sign of the z-statistics does not change across stages (faster,

	recommended when many interim analyses, e.g., more than 10, are performed). Region probabilities are numerically integrated under both settings. Defaults to TRUE
integer	Logical indicating whether only integer-valued maximum sample sizes should be returned. If TRUE, the required maximum sample size is rounded to the next larger integer. Defaults to TRUE.
search	Sample-size search rule for schedules generated from looks/timing. "adaptive" uses bracketing and binary search with local checks. It is a heuristic that can skip an isolated crossing on a non-monotone power curve, and stops with a diagnostic if a transient invalid candidate invalidates the bracket. Use "exhaustive" when the first crossing must be guaranteed. "exhaustive" scans the full feasible candidate maximum-sample-size range and returns the first finite candidate that reaches power, skipping transient invalid candidates and stopping at terminal invalid candidates. Ignored when by is supplied, where scheduled maximum sample sizes minN + j * by are scanned in increasing order. Defaults to "adaptive".
details	Logical indicating whether the full search result should be returned instead of only the maximum sample size. The detailed result is scalar; vectorized inputs require details = FALSE. Defaults to FALSE.
alternative	Direction of the alternative hypothesis, one of "two.sided" (default), "less", or "greater". For normal and moment priors, one-sided alternatives truncate and renormalize the analysis prior below or above null, respectively, while retaining the point null hypothesis. Prior parameters describe the distribution before truncation. A normal point prior (psd = 0) must lie strictly on the specified side of null.
...	Additional arguments passed to pbf01seq.

Details

The function searches over the maximum sample size of the sequential design. Candidate look schedules are rebuilt for each maximum sample size according to looks/timing or by/minN. Power need not be monotone for every design-prior and target combination, and rounded interim looks add further non-monotonicity; search selects the search rule. For by/minN schedules, scheduled maximum sample sizes are scanned in increasing order and previous look calculations are reused; search is ignored. If the target is not reached within nrange, the function returns NaN and issues a warning.

Value

The maximum sample size found to achieve the specified power. If details = TRUE, returns a list with the sample size, achieved power, generated design, and search diagnostics.

Author(s)

František Bartoš

See Also

pbf01seq, powerbf01seq, nbf01

Examples

```
nbf01seq(k1 = 1/5, k0 = 5, power = 0.8, usd = sqrt(2),
         pm = 0, psd = 1, dpm = 0.5, dpsd = 0,
         looks = 3, nrange = c(2, 200))
```

---

nbinbf01 *Sample size determination for binomial Bayes factor*

Description

This function computes the required sample size to obtain a binomial Bayes factor (binbf01) more extreme than a threshold k with a specified target power.

Usage

```
nbinbf01(
  k,
  power,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1,
  lower.tail = TRUE,
  nrange = c(1, 10^4),
  ...
)
```

Arguments

k	Bayes factor threshold
power	Target power
p0	Tested binomial proportion. Defaults to 0.5
type	Type of test. Can be "point" or "direction". Defaults to "point"
a	Number of successes parameter of the beta analysis prior distribution. Defaults to 1
b	Number of failures parameter of the beta analysis prior distribution. Defaults to 1
dp	Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA

da	Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior
db	Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior
dl	Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0
du	Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1
lower.tail	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq k)$ (TRUE) or $\Pr(\mathrm{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE
nrange	Sample size search range over which numerical search is performed. Defaults to c(1, 10^4)
...	Other arguments passed to stats::uniroot. The sample-size root tolerance defaults to tol = 1e-8 and can be overridden.

Value

The required sample size to achieve the specified power

Author(s)

Samuel Pawel

See Also

pbinbf01, binbf01

Examples

```
## sample size parameters
pow <- 0.9
p0 <- 3/4
a <- 1
b <- 1
k <- 1/10

## Not run:
## sample sizes for directional testing
(nH1 <- nbinbf01(k = k, power = pow, p0 = p0, type = "direction", a = a,
                 b = b, da = a, db = b, dl = p0, du = 1))
(nH0 <- nbinbf01(k = 1/k, power = pow, p0 = p0, type = "direction", a = a,
                 b = b, da = a, db = b, dl = 0, du = p0, lower.tail = FALSE))
nseq <- seq(1, 1.1*max(c(nH1, nH0)), length.out = 100)
powH1 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "direction", a = a,
                  b = b, da = a, db = b, dl = p0, du = 1)
powH0 <- pbinbf01(k = 1/k, n = nseq, p0 = p0, type = "direction", a = a,
                  b = b, da = a, db = b, dl = 0, du = p0, lower.tail = FALSE)
```

```
matplot(nseq, cbind(powH1, powH0), type = "s", xlab = "n", ylab = "Power", lty = 1,
        ylim = c(0, 1), col = c(2, 4), las = 1)
abline(h = pow, lty = 2)
abline(v = c(nH1, nH0), col = c(2, 4), lty = 2)
legend("topleft", legend = c("H1", "H0"), lty = 1, col = c(2, 4))

## sample sizes for point null testing
(nH1 <- nbinbf01(k = k, power = pow, p0 = p0, type = "point", a = a,
                 b = b, da = a, db = b))
(nH0 <- nbinbf01(k = 1/k, power = pow, p0 = p0, type = "point", a = a,
                 b = b, dp = p0, lower.tail = FALSE, nrange = c(1, 10^5)))
nseq <- seq(1, max(c(nH1, nH0)), length.out = 100)
powH1 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "point", a = a,
                  b = b, da = a, db = b, dl = 0, du = 1)
powH0 <- pbinbf01(k = 1/k, n = nseq, p0 = p0, type = "point", a = a,
                  b = b, dp = p0, lower.tail = FALSE)
matplot(nseq, cbind(powH1, powH0), type = "s", xlab = "n", ylab = "Power", lty = 1,
        ylim = c(0, 1), col = c(2, 4), las = 1)
abline(h = pow, lty = 2)
abline(v = c(nH1, nH0), col = c(2, 4), lty = 2)
legend("topleft", legend = c("H1", "H0"), lty = 1, col = c(2, 4))

## End(Not run)
```

---

nmbf01 *Normal moment prior Bayes factor*

Description

This function computes the Bayes factor that quantifies the evidence that the data (in the form of an asymptotically normally distributed parameter estimate with standard error) provide for a point null hypothesis with a normal moment prior assigned to the parameter under the alternative.

Usage

```
nmbf01(
  estimate,
  se,
  null = 0,
  psd,
  log = FALSE,
  alternative = c("two.sided", "less", "greater")
)
```

Arguments

estimate	Parameter estimate
se	Standard error of the parameter estimate

`null`	Parameter value under the point null hypothesis. Defaults to `0`
`psd`	Spread of the normal moment prior assigned to the parameter under the alternative. The modes of the prior are located at $\pm\sqrt{2}$ `psd`
`log`	Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to `FALSE`
`alternative`	Direction of the alternative hypothesis, one of `"two.sided"` (default), `"less"`, or `"greater"`. For a one-sided alternative, the normal moment analysis prior is restricted to values below `null` (`"less"`) or above `null` (`"greater"`) and renormalized. The point null hypothesis is unchanged; `psd` is the scale before truncation.

Details

A normal moment prior has density $f(x \mid \texttt{null}, \texttt{psd}) = N(x \mid \texttt{null}, \texttt{psd}^2) \times (x - \texttt{null})^2/\texttt{psd}^2$ with $N(x \mid m, v)$ the normal density with mean m and variance v evaluated at x.

Value

Bayes factor in favor of the null hypothesis over the alternative ($BF_{01} > 1$ indicates evidence for the null hypothesis, whereas $BF_{01} < 1$ indicates evidence for the alternative)

Author(s)

Samuel Pawel

References

Johnson, V. E. and Rossell, D. (2010). On the use of non-local prior densities in Bayesian hypothesis tests. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 72(2):143–170. doi:10.1111/j.14679868.2009.00730.x

Pramanik, S. and Johnson, V. E. (2024). Efficient alternatives for Bayesian hypothesis tests in psychology. Psychological Methods, 29(2):243–261. doi:10.1037/met0000482

See Also

nmbf01, pnmbf01, nnmbf01, powernmbf01

Examples

```
nmbf01(estimate = 0.25, se = 0.05, null = 0, psd = 0.5/sqrt(2)) # mode at 0.5

## positive normal moment alternative
nmbf01(estimate = 0.25, se = 0.05, psd = 0.5/sqrt(2), alternative = "greater")
```

nnmbf01 *Sample size determination for normal moment prior Bayes factor*

Description

This function computes the required sample size to obtain a normal moment prior Bayes factor (nbf01) more extreme than a threshold k with a specified target power.

Usage

```
nnmbf01(
  k,
  power,
  usd,
  null = 0,
  psd,
  dpm,
  dpsd,
  nrange = c(1, 10^5),
  lower.tail = TRUE,
  integer = TRUE,
  ...,
  alternative = c("two.sided", "less", "greater")
)
```

Arguments

k	Bayes factor threshold
power	Target power
usd	Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$, see details
null	Parameter value under the point null hypothesis. Defaults to 0
psd	Spread of the normal moment prior assigned to the parameter under the alternative in the analysis. The modes of the prior are located at $\pm\sqrt{2}\,$psd
dpm	Mean of the normal design prior assigned to the parameter
dpsd	Standard deviation of the normal design prior assigned to the parameter. Set to 0 to obtain a point prior at the design prior mean
nrange	Sample size search range over which numerical search is performed. Defaults to c(1, 10^5)
lower.tail	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq k)$ (TRUE) or $\Pr(\mathrm{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE
integer	Logical indicating whether only integer valued sample sizes should be returned. If TRUE the required sample size is rounded to the next larger integer. Defaults to TRUE

`...` Other arguments passed to `stats::uniroot`. The sample-size root tolerance defaults to `tol = 1e-8` and can be overridden.

`alternative` Direction of the alternative hypothesis, one of `"two.sided"` (default), `"less"`, or `"greater"`. For a one-sided alternative, the normal moment analysis prior is restricted to values below `null` (`"less"`) or above `null` (`"greater"`) and renormalized. The point null hypothesis is unchanged; `psd` is the scale before truncation. Must be specified by name: unnamed arguments after `integer` are passed to `stats::uniroot`.

Details

It is assumed that the standard error of the future parameter estimate is of the form $\mathtt{se} = \mathtt{usd}/\sqrt{\mathtt{n}}$. For example, for normally distributed data with known standard deviation `sd` and two equally sized groups of size `n`, the standard error of an estimated standardized mean difference is $\mathtt{se} = \mathtt{sd}\sqrt{2/n}$, so the corresponding unit standard deviation is $\mathtt{usd} = \mathtt{sd}\sqrt{2}$. See the vignette for more information.

Value

The required sample size to achieve the specified power

Note

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors. The normal design prior remains untruncated for one-sided alternatives. Power need not increase with sample size; for example, the probability of misleading evidence first rises and then falls. If both endpoints of `nrange` are below the target, the numerical search checks the interior maximum. The returned sample size is the smallest at which power reaches the target; power may fall below the target again at larger sample sizes.

Author(s)

Samuel Pawel

See Also

nmbf01, pnmbf01, powernmbf01

Examples

```
nnmbf01(k = 1/10, power = 0.9, usd = 1, null = 0, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0)
```

ntbf01 *Sample size calculations for t-test Bayes factor*

Description

This function computes the required sample size to obtain a t-test Bayes factor (tbf01) more extreme than a threshold k with a specified target power.

Usage

```
ntbf01(
  k,
  power,
  null = 0,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  dpm = plocation,
  dpsd = pscale,
  lower.tail = TRUE,
  integer = TRUE,
  nrange = c(2, 10^4),
  ratio = 1,
  drange = "adaptive",
  tail.eps = getOption("bfpwr.tail.eps", 1e-06),
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  ...
)
```

Arguments

k	Bayes factor threshold
power	Target power
null	Standardized mean difference under the point null hypothesis. Defaults to 0
plocation	t prior location. Defaults to 0
pscale	t prior scale. Defaults to 1/sqrt(2)
pdf	t prior degrees of freedom. Defaults to 1 (a Cauchy prior)
type	Type of t-test. Can be "two.sample" (default), "one.sample", or "paired"
alternative	Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to effects below and above null, respectively.
dpm	Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location

dpsd	Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale
lower.tail	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq k)$ (TRUE) or $\Pr(\mathrm{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE
integer	Logical indicating whether only integer valued sample sizes should be returned. If TRUE the required sample size is rounded to the next larger integer. Defaults to TRUE
nrange	Sample size search range over which numerical search is performed. Defaults to c(2, 10^4)
ratio	Allocation ratio n2 / n1 for two-sample designs. Candidate group-2 sample sizes are ceiling(n1 * ratio). Ignored for one-sample and paired designs.
drange	Standardized mean difference search range over which the critical values are searched for. Can be either set to a numerical range or to "adaptive" (default) which determines the range in an adaptive way from the other input parameters
tail.eps	One-sided adaptive power-boundary searches stop once the remaining predictive probability in the searched tail is at most this value. If a fixed-n power evaluation reaches that cutoff before finding a boundary, the returned 0/1 tail probability has omitted mass bounded by tail.eps. Smaller values search farther. Defaults to 1e-6
tail.nquad	Number of Gauss-Legendre quadrature nodes used by tbf01 for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to 512.
...	Optional numerical controls passed to ptbf01 and to the stats::uniroot sample-size search. The fixed-design critical-value range drange is handled explicitly and is not passed to the sample-size search.

Value

The required sample size to achieve the specified power

Note

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors. The normal design prior remains untruncated for one-sided alternatives. Power need not increase with sample size; for example, the probability of misleading evidence first rises and then falls. If both endpoints of nrange are below the target, the numerical search checks the interior maximum. The returned sample size is the smallest at which power reaches the target; power may fall below the target again at larger sample sizes.

Author(s)

Samuel Pawel, František Bartoš

See Also

ptbf01, powertbf01, tbf01

Examples

```
## example from Schönbrodt and Wagenmakers (2018, p.135)
ntbf01(k = 1/6, power = 0.95, dpm = 0.5, dpsd = 0, alternative = "greater")
ntbf01(k = 1/6, power = 0.95, dpm = 0.5, dpsd = 0.1, alternative = "greater")
ntbf01(k = 6, power = 0.95, dpm = 0, dpsd = 0, alternative = "greater",
       lower.tail = FALSE, nrange = c(2, 10000))
```

`ntbf01seq` *Maximum Sample Size Determination for Sequential t-Test Bayes Factors*

Description

Computes the maximum sample size required for a sequential t-test Bayes factor design to reach a target probability of stopping for H_1 or H_0.

Usage

```
ntbf01seq(
  k1,
  k0 = 1/k1,
  power,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  dpm = plocation,
  dpsd = pscale,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  target = c("H1", "H0"),
  nrange = c(2, 10^4),
  looks = 1,
  timing = NULL,
  minN = NULL,
  by = NULL,
  ratio = 1,
  strict = TRUE,
  trange = "adaptive",
  tail.eps = getOption("bfpwr.tail.eps", 1e-06),
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  integer = TRUE,
  search = c("adaptive", "exhaustive"),
  details = FALSE,
  ...
)
```

Arguments

k1	Bayes factor threshold in favor of H_1. Evidence for H_1 is obtained when $\mathrm{BF}_{01} \leq k1$.
k0	Bayes factor threshold in favor of H_0. Evidence for H_0 is obtained when $\mathrm{BF}_{01} \geq k0$.
power	Target stopping probability.
plocation	t prior location. Defaults to `0`
pscale	t prior scale. Defaults to `1/sqrt(2)`
pdf	t prior degrees of freedom. Defaults to `1` (a Cauchy prior)
dpm	Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location
dpsd	Standard deviation of the normal design prior assigned to the standardized mean difference. Set to `0` to obtain a point prior at the design prior mean. Defaults to the analysis prior scale
type	Type of t-test. Can be `"two.sample"` (default), `"one.sample"`, or `"paired"`
alternative	Direction of the test. Can be either `"two.sided"` (default), `"less"`, or `"greater"`. The latter two truncate the analysis prior to effects below and above `null`, respectively.
target	Character string. Either `"H1"` for the final cumulative probability of stopping for H_1, or `"H0"` for the final cumulative probability of stopping for H_0.
nrange	Maximum sample size search range over which numerical search is performed. Defaults to `c(2, 10^4)`.
looks	Number of sequential looks when `timing` is not supplied. The default `looks = 1` gives a fixed design.
timing	Optional cumulative information fractions for the looks. Values must be positive, strictly increasing, no larger than one, and end at one. Candidate look sample sizes are `ceiling(maximumN * timing - sqrt(.Machine$double.eps))` with the final look fixed to `maximumN`.
minN	First-look sample size when `by` is supplied. Defaults to the lower search bound.
by	Optional sample-size increase between looks. If supplied, schedules are generated as `seq(minN, maximumN, by = by)` with `maximumN` appended as the final look.
ratio	Allocation ratio `n2 / n1` for two-sample designs. Candidate group-2 sample sizes are `ceiling(n1 * ratio)`. Ignored for one-sample and paired designs.
strict	Logical. If `TRUE` and there are more than two critical values per stage, enumerate all possible region combinations. If `FALSE`, only integrate over the main regions where the sign of the z-statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Region probabilities are numerically integrated under both settings. Defaults to `TRUE`
trange	Critical t-statistic search strategy for the sequential stopping boundaries. Can be either `"adaptive"` (default) or a numeric interval. For one-sided adaptive searches, `tail.eps` determines the predictive tail cutoff used to stop finite scans. Use a numeric interval to force exact boundary searches over that interval.

`tail.eps` One-sided adaptive sequential boundary searches stop once the marginal predictive tail probability beyond the searched t statistic is at most `tail.eps`. The same cutoff is used for each adaptive boundary search; it is not split across looks or boundaries. If no boundary is found before the cutoff, it is placed at infinity in the searched direction, preserving the H0 and H1 sides of the stopping event. The omitted marginal mass is bounded by `tail.eps`; with C unresolved boundary searches, a conservative combined bound is `min(1, C * tail.eps)` rather than `tail.eps`. smaller values search farther at additional computational cost. Defaults to `1e-6`.

`tail.nquad` Number of Gauss-Legendre quadrature nodes used by `tbf01` for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to `512`.

`integer` Logical indicating whether only integer-valued maximum sample sizes should be returned. If `TRUE`, the required maximum sample size is rounded to the next larger integer. Defaults to `TRUE`.

`search` Sample-size search rule for schedules generated from `looks`/`timing`. `"adaptive"` uses bracketing and binary search with local checks. It is a heuristic that can skip an isolated crossing on a non-monotone power curve, and stops with a diagnostic if a transient invalid candidate invalidates the bracket. Use `"exhaustive"` when the first crossing must be guaranteed. `"exhaustive"` scans the full feasible candidate maximum-sample-size range and returns the first finite candidate that reaches `power`, skipping transient invalid candidates and stopping at terminal invalid candidates. Ignored when `by` is supplied, where scheduled maximum sample sizes `minN + j * by` are scanned in increasing order. Defaults to `"adaptive"`.

`details` Logical indicating whether the full search result should be returned instead of only the maximum sample size. The detailed result is scalar; vectorized inputs require `details = FALSE`. Defaults to `FALSE`.

`...` Additional arguments passed to `ptbf01seq`.

Details

The function searches over the maximum group-1 sample size for two-sample designs, or the maximum sample size for one-sample and paired designs. Candidate look schedules are rebuilt for each maximum sample size according to `looks`/`timing` or `by`/`minN`. Power need not be monotone for every design-prior and target combination, and rounded interim looks add further non-monotonicity; `search` selects the search rule. For `by`/`minN` schedules, scheduled maximum sample sizes are scanned in increasing order and previous look calculations are reused; `search` is ignored. If the target is not reached within `nrange`, the function returns `NaN` and issues a warning.

Value

The maximum sample size in group 1 found to achieve the specified power. If `details = TRUE`, returns a list with the sample size, achieved power, generated design, and search diagnostics.

Author(s)

František Bartoš

See Also

ptbf01seq, powertbf01seq, ntbf01

Examples

```
ntbf01seq(k1 = 1/2, k0 = 2, power = 0.4, dpm = 0.5, dpsd = 0,
          alternative = "greater", looks = 2, nrange = c(2, 80),
          strict = FALSE)
```

---

pbf01 *Cumulative distribution function of the z-test Bayes factor*

Description

This function computes the probability of obtaining a Bayes factor (bf01) more extreme than a threshold k with a specified sample size.

Usage

```
pbf01(
  k,
  n,
  usd,
  null = 0,
  pm,
  psd,
  dpm = pm,
  dpsd = psd,
  lower.tail = TRUE,
  alternative = c("two.sided", "less", "greater")
)
```

Arguments

k	Bayes factor threshold
n	Sample size
usd	Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$, see details
null	Parameter value under the point null hypothesis. Defaults to 0
pm	Mean of the normal prior assigned to the parameter under the alternative in the analysis
psd	Standard deviation of the normal prior assigned to the parameter under the alternative in the analysis. Set to 0 to obtain a point prior at the prior mean
dpm	Mean of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior pm

dpsd	Standard deviation of the normal design prior assigned to the parameter. Defaults to the same value as the analysis prior psd
lower.tail	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq \mathsf{k})$ (TRUE) or $\Pr(\mathrm{BF}_{01} > \mathsf{k})$ (FALSE) should be computed. Defaults to TRUE
alternative	Direction of the alternative hypothesis, one of "two.sided" (default), "less", or "greater". For a one-sided alternative, the normal analysis prior is restricted to values below null ("less") or above null ("greater") and renormalized. pm and psd describe the normal distribution before truncation. A point prior (psd = 0) must lie strictly on the specified side of null.

Details

It is assumed that the standard error of the future parameter estimate is of the form $\mathsf{se} = \mathsf{usd}/\sqrt{\mathsf{n}}$. For example, for normally distributed data with known standard deviation sd and two equally sized groups of size n, the standard error of an estimated standardized mean difference is $\mathsf{se} = \mathsf{sd}\sqrt{2/n}$, so the corresponding unit standard deviation is $\mathsf{usd} = \mathsf{sd}\sqrt{2}$. See the vignette for more information.

Value

The probability that the Bayes factor is less or greater (depending on the specified lower.tail) than the specified threshold k

Note

For a one-sided alternative, the normal design prior remains untruncated; truncation applies only to the analysis prior.

Author(s)

Samuel Pawel, František Bartoš

See Also

nbf01, powerbf01, bf01

Examples

```
## point alternative (psd = 0)
pbf01(k = 1/10, n = 200, usd = 2, null = 0, pm = 0.5, psd = 0)

## normal alternative (psd > 0)
pbf01(k = 1/10, n = 100, usd = 2, null = 0, pm = 0.5, psd = 2)

## design prior is the null hypothesis (dpm = 0, dpsd = 0)
pbf01(k = 10, n = 1000, usd = 2, null = 0, pm = 0.3, psd = 2, dpm = 0, dpsd = 0, lower.tail = FALSE)

## draw a power curve
nseq <- round(exp(seq(log(10), log(10000), length.out = 100)))
plot(nseq, pbf01(k = 1/10, n = nseq, usd = 2, null = 0, pm = 0.3, psd = 0), type = "l",
     xlab = "n", ylab = bquote("Pr(BF"["01"] <= 1/10 * ")"), ylim = c(0, 1),
```

```
     log = "x", las = 1)

## standardized mean difference (usd = sqrt(2), effective sample size = per group size)
n <- 30
pbf01(k = 1/10, n = n, usd = sqrt(2), null = 0, pm = 0, psd = 1)

## z-transformed correlation (usd = 1, effective sample size = n - 3)
n <- 100
pbf01(k = 1/10, n = n - 3, usd = 1, null = 0, pm = 0.2, psd = 0.5)

## log hazard/odds ratio (usd = 2, effective sample size = total number of events)
nevents <- 100
pbf01(k = 1/10, n = nevents, usd = 2, null = 0, pm = 0, psd = sqrt(0.5))
```

`pbf01seq` *Sequential Bayes Factor Design*

Description

Computes cumulative probabilities of observing z-test Bayes factors that provide evidence for the null hypothesis H_0, the alternative hypothesis H_1, or remain inconclusive in a sequential design. Optionally, also computes the expected sample size.

Usage

```
pbf01seq(
  k1,
  k0 = 1/k1,
  se,
  n = NULL,
  null = 0,
  pm = NULL,
  psd,
  dpm = pm,
  dpsd = psd,
  type = c("normal", "directional", "moment"),
  strict = TRUE,
  alternative = c("two.sided", "less", "greater"),
  ...
)
```

Arguments

`k1`	Bayes factor threshold in favor of H_1 (i.e., $\text{BF}_{01} \leq$ `k1` < 1 implies evidence for H_1)
`k0`	Bayes factor threshold in favor of H_0 (i.e., $\text{BF}_{01} \geq$ `k0` > 1 implies evidence for H_0)

se	Numeric vector of standard errors for each sequential stage
n	Optional numeric vector of sample sizes corresponding to se. If supplied, the expected sample size is computed
null	Parameter value under the point null hypothesis. Defaults to 0
pm	Analysis prior mean. Not taken into account for type = "moment" for the analysis prior. If dpm is omitted, its default remains pm, including for moment priors
psd	Analysis prior standard deviation (type = "normal" and type = "directional") or scale (type = "moment")
dpm	Mean of the normal design prior
dpsd	Standard deviation of the normal design prior. Set dpsd = 0 to obtain a point prior at dpm
type	Character string. One of • "normal" (default): point null vs. normal alternative (set psd = 0 to obtain a point alternative) • "directional": directional null $\theta \leq$ null vs. directional alternative $\theta >$ null with a marginal normal prior • "moment": point null vs. normal moment alternative which is centered around null One-sided alternative values are supported for type = "normal" and type = "moment"; type = "directional" compares two composite hypotheses and retains its existing meaning.
strict	Logical. If TRUE and there are more than two critical values per stage, enumerate all possible region combinations. If FALSE, only integrate over the main regions where the sign of the z-statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Region probabilities are numerically integrated under both settings. Defaults to TRUE
alternative	Direction of the alternative hypothesis, one of "two.sided" (default), "less", or "greater". For normal and moment priors, one-sided alternatives truncate and renormalize the analysis prior below or above null, respectively, while retaining the point null hypothesis. Prior parameters describe the distribution before truncation. A normal point prior (psd = 0) must lie strictly on the specified side of null.
...	Numerical integration controls. Use ngrid to set the number of deterministic Halton grid points used by mvtnorm::lpmvnorm (default 10000). Alternatively, set method = "pmvnorm" and pass controls for mvtnorm::pmvnorm. Session defaults can be changed with bfpwrOptions; explicit controls take precedence.

Details

The function constructs per-stage integration regions for cumulative z-statistics based on the Bayes factor thresholds k1 and k0, then computes the probability of these regions under a predictive distribution defined by se and the normal design prior with dpm and dpsd. Integration is performed via mvtnorm::lpmvnorm. This is deterministic numerical integration; increasing ngrid can be used

to check convergence. Very small stopping probabilities can be missed by the integration grid; increasing its size does not guarantee relative accuracy for rare events. The null value and the analysis and design prior means are all specified on the original parameter scale, matching the fixed-sample z-test functions. The normal design prior is not truncated, including for one-sided analysis priors. One-sided Bayes factor boundaries are found numerically using the same calculation as pbf01 or pnmbf01.

Value

An object of class "bfseqdesign", which is a list containing the input arguments, the critical z-values, the expected sample size, the cumulative probabilities of stopping for H_1 and H_0 by each stage, and the cumulative probabilities of remaining inconclusive by each stage.

Author(s)

Samuel Pawel, František Bartoš

Examples

```
n <- seq(50, 200, 50) # sample size per stage
se <- sqrt(2/n) # standard errors per stage
res <- pbf01seq(k1 = 1/10, k0 = 3, se = se, n = n, pm = 0, psd = 1,
                dpm = 0.5, dpsd = 0.05, type = "normal")
res # print summary
plot(res) # plot summary
res$cumpH1 # cumulative probability to stop for H1 by each stage
res$cumpH0 # cumulative probability to stop for H0 by each stage
res$EN # expected sample size
```

---

pbinbf01 *Cumulative distribution function of the binomial Bayes factor*

Description

This function computes the probability of obtaining a binomial Bayes factor (binbf01) more extreme than a threshold k with a specified sample size.

Usage

```
pbinbf01(
  k,
  n,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  dp = NA,
```

```
  da = a,
  db = b,
  dl = 0,
  du = 1,
  lower.tail = TRUE
)
```

Arguments

k	Bayes factor threshold
n	Number of trials
p0	Tested binomial proportion. Defaults to 0.5
type	Type of test. Can be "point" or "direction". Defaults to "point"
a	Number of successes parameter of the beta analysis prior distribution. Defaults to 1
b	Number of failures parameter of the beta analysis prior distribution. Defaults to 1
dp	Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA
da	Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior
db	Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior
dl	Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0
du	Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1
lower.tail	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq \mathrm{k})$ (TRUE) or $\Pr(\mathrm{BF}_{01} > \mathrm{k})$ (FALSE) should be computed. Defaults to TRUE

Value

The probability that the Bayes factor is less or greater (depending on the specified lower.tail) than the specified threshold k

Author(s)

Samuel Pawel

See Also

binbf01, nbinbf01

Examples

```
## compute probability that BF > 10 under the point null
a <- 1
b <- 1
p0 <- 3/4
k <- 10
nseq <- seq(1, 1000, length.out = 50)
powH0 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "point", a = a, b = b,
                  dp = p0, lower.tail = FALSE)
plot(nseq, powH0, type = "s", xlab = "n", ylab = "Power")

## compare to normal approximation
pm <- a/(a + b) # prior mean under H1
psd <- sqrt(a*b/(a + b)^2/(a + b + 1)) # prior standard deviation under H1
pownormH0 <- pbf01(k = k, n = nseq, usd = sqrt(p0*(1 - p0)), null = p0,
                   pm = pm, psd = psd, dpm = p0, dpsd = 0, lower.tail = FALSE)
lines(nseq, pownormH0, type = "s", col = 2)
legend("right", legend = c("Exact", "Normal approximation"), lty = 1,
       col = c(1, 2))

## compute probability that BF < 1/10 under the p|H1 ~ Beta(a, b) alternative
a <- 10
b <- 5
p0 <- 3/4
k <- 1/10
powH1 <- pbinbf01(k = k, n = nseq, p0 = p0, type = "point", a = a, b = b,
                  da = a, db = b, dl = 0, du = 1)
plot(nseq, powH1, type = "s", xlab = "n", ylab = "Power")

## compare to normal approximation
pm <- a/(a + b) # prior mean under H1
psd <- sqrt(a*b/(a + b)^2/(a + b + 1)) # prior standard deviation under H1
pownormH1 <- pbf01(k = k, n = nseq, usd = sqrt(pm*(1 - pm)), null = p0,
                   pm = pm, psd = psd, dpm = pm, dpsd = psd)
lines(nseq, pownormH1, type = "s", col = 2)
legend("right", legend = c("Exact", "Normal approximation"), lty = 1,
       col = c(1, 2))

## probability that directional BF <= 1/10 under uniform [3/4, 1] design prior
pow <- pbinbf01(k = 1/10, n = nseq, p0 = 3/4, type = "direction", a = 1, b = 1,
                da = 1, db = 1, dl = 3/4, du = 1)
plot(nseq, pow, type = "s", xlab = "n", ylab = "Power")
```

`pbinbf01seq` *Power calculations for binomial Bayes factor with two-stage designs*

Description

Compute probability that binomial Bayes factor (binbf01) is equal or below a threshold at final stage, taking into account that the study can stop for futility if it is greater than a specified threshold at interim.

Usage

```
pbinbf01seq(
  n1,
  n2,
  k = 1/10,
  kf = 1/k,
  p0 = 0.5,
  type = c("direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1
)
```

Arguments

n1	Sample size at interim stage
n2	Sample size at final stage
k	Bayes factor threshold for evidence for efficacy. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis
kf	Bayes factor threshold for evidence for futility. Defaults to 1/k.
p0	Tested binomial proportion. Defaults to 0.5
type	Type of test. Currently only "direction" is implemented
a	Number of successes parameter of the beta analysis prior distribution. Defaults to 1
b	Number of failures parameter of the beta analysis prior distribution. Defaults to 1
dp	Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA
da	Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior
db	Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior
dl	Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0
du	Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1

Value

The probability of obtaining a Bayes factor equal or below k at the final stage taking into account potential futility stopping with BF01 > kf at interim

Author(s)

Samuel Pawel

See Also

binbf01, powerbinbf01seq

Examples

```
## probability of BF<1/10 under alternative truncated beta design prior
pbinbf01seq(n1 = 10, n2 = 20, k = 1/10, kf = 3, p0 = 0.2, a = 1, b = 1,
            dl = 0.2, du = 1, type = "direction") # with interim analysis
pbinbf01(n = 20, k = 1/10, p0 = 0.2, a = 1, b = 1, dl = 0.2, du = 1,
         type = "direction") # without interim analysis

## probability to stop at interim under null truncated beta design prior
pbinbf01(n = 10, k = 3, p0 = 0.2, a = 1, b = 1, dl = 0, du = 0.2,
         type = "direction", lower.tail = FALSE)

## probability of BF<1/10 under point null design prior
pbinbf01seq(n1 = 10, n2 = 20, k = 1/10, kf = 3, p0 = 0.2, a = 1, b = 1,
            dp = 0.2, type = "direction") # with interim analysis
pbinbf01(n = 20, k = 1/10, p0 = 0.2, a = 1, b = 1, dp = 0.2,
         type = "direction") # without interim analysis

## probability to stop at interim under point null design prior
pbinbf01(n = 10, k = 3, p0 = 0.2, a = 1, b = 1, dp = 0.2,
         type = "direction", lower.tail = FALSE)
```

---

plot.bfseqdesign *Plot method for class* "bfseqdesign"

Description

Plot method for class "bfseqdesign"

Usage

```
## S3 method for class 'bfseqdesign'
plot(
  x,
  plot = TRUE,
  nullplot = TRUE,
```

```
  zplot = FALSE,
  digits = max(3L, getOption("digits") - 3L),
  ...
)
```

Arguments

x	Object of class "power.bftest"
plot	Logical indicating whether data should be plotted. If FALSE only the data used for plotting are returned.
nullplot	Logcal indicating whether a second plot with the stopping probabilities computed under the null hypothesis should also be produced. Defaults to TRUE
zplot	Logcal indicating whether a plot of the critical z-values should be produced
digits	Number of digits for formatting of numbers
...	Other arguments (for consistency with the generic)

Value

Plots stopping curves (if specified) and invisibly returns a list of data frames containing the data underlying the stopping curves

Author(s)

Samuel Pawel, František Bartoš

See Also

pbf01seq

Examples

```
n <- seq(25, 150, 25) # sample size per stage
se <- sqrt(2/n) # standard errors per stage
res <- pbf01seq(k1 = 1/10, k0 = 3, se = se, n = n, pm = 0, psd = 1,
                dpm = 0.5, dpsd = 0.1, type = "moment")
plot(res, nullplot = FALSE) # only under design prior
plot(res, nullplot = TRUE) # also plot under null hypothesis
plot(res, zplot = TRUE) # show critical z-values

## case which causes trouble
pm <- 0
psd <- 5
dpm <- 24
dpsd <- 0
seseq <- c(12.030479, 8.506833, 6.945800, 6.015239, 5.380194, 4.911422, 4.547094,
           4.253417, 4.010160, 3.804371)
nseq <- seq(2, 20, 2)
pbf01seq(k1 = 1/10, k0 = 3, se = seseq, n = nseq, pm = pm, psd = psd,
         dpm = dpm, dpsd = dpsd, type = "directional")
```

`plot.power.bftest` *Plot method for class* `"power.bftest"`

Description

Plot method for class `"power.bftest"`

Usage

```
## S3 method for class 'power.bftest'
plot(
  x,
  nlim = c(2, 500),
  ngrid = 100,
  type = "l",
  plot = TRUE,
  nullplot = TRUE,
  ...
)
```

Arguments

`x`	Object of class `"power.bftest"`
`nlim`	Range of sample sizes over which the power should be computed. Defaults to `c(2, 500)`
`ngrid`	Number of grid point for which power should be computed. Defaults to 100
`type`	Type of plot. Defaults to `"l"` (line-plot)
`plot`	Logical indicating whether data should be plotted. If `FALSE` only the data used for plotting are returned.
`nullplot`	Logcal indicating whether a second plot with the power in favor of the null (using a Bayes factor threshold of 1/k) should be created. Defaults to `TRUE`
`...`	Other arguments (for consistency with the generic)

Value

Plots power curves (if specified) and invisibly returns a list of data frames containing the data underlying the power curves

Author(s)

Samuel Pawel, František Bartoš

See Also

powerbf01, powertbf01, powernmbf01

Examples

```
ssd1 <- powerbf01(k = 1/6, power = 0.95, pm = 0, psd = 1/sqrt(2), dpm = 0.5, dpsd = 0)
plot(ssd1, nlim = c(1, 8000))

power1 <- powerbf01(k = 1/2, n = 120, pm = 0, psd = 1/sqrt(2), dpm = 0.5, dpsd = 0)
plot(power1, nlim = c(1, 1000))
```

---

pnmbf01 *Cumulative distribution function of the normal moment prior Bayes factor*

Description

This function computes the probability of obtaining a normal moment prior Bayes factor (nmbf01) more extreme than a threshold k with a specified sample size.

Usage

```
pnmbf01(
  k,
  n,
  usd,
  null = 0,
  psd,
  dpm,
  dpsd,
  lower.tail = TRUE,
  alternative = c("two.sided", "less", "greater")
)
```

Arguments

k	Bayes factor threshold
n	Sample size
usd	Unit standard deviation, the (approximate) standard error of the parameter estimate based on $n = 1$, see details
null	Parameter value under the point null hypothesis. Defaults to 0
psd	Spread of the normal moment prior assigned to the parameter under the alternative in the analysis. The modes of the prior are located at $\pm\sqrt{2}\,$psd
dpm	Mean of the normal design prior assigned to the parameter
dpsd	Standard deviation of the normal design prior assigned to the parameter. Set to 0 to obtain a point prior at the design prior mean
lower.tail	Logical indicating whether $\Pr(\mathrm{BF}_{01} \leq k)$ (TRUE) or $\Pr(\mathrm{BF}_{01} > k)$ (FALSE) should be computed. Defaults to TRUE

`alternative` Direction of the alternative hypothesis, one of `"two.sided"` (default), `"less"`, or `"greater"`. For a one-sided alternative, the normal moment analysis prior is restricted to values below `null` (`"less"`) or above `null` (`"greater"`) and renormalized. The point null hypothesis is unchanged; `psd` is the scale before truncation.

Details

It is assumed that the standard error of the future parameter estimate is of the form $\mathsf{se} = \mathsf{usd}/\sqrt{\mathsf{n}}$. For example, for normally distributed data with known standard deviation `sd` and two equally sized groups of size `n`, the standard error of an estimated standardized mean difference is $\mathsf{se} = \mathsf{sd}\sqrt{2/n}$, so the corresponding unit standard deviation is $\mathsf{usd} = \mathsf{sd}\sqrt{2}$. See the vignette for more information.

Value

The probability that the Bayes factor is less or greater (depending on the specified `lower.tail`) than the specified threshold `k`

Note

The normal design prior is not truncated, including for one-sided analysis priors.

Author(s)

Samuel Pawel

See Also

nmbf01, nnmbf01, powernmbf01

Examples

```
## point design prior (dpsd = 0)
pnmbf01(k = 1/10, n = 200, usd = 2, null = 0, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0)

## normal design prior to incorporate parameter uncertainty (dpsd > 0)
pnmbf01(k = 1/10, n = 200, usd = 2, null = 0, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0.25)

## design prior is the null hypothesis (dpm = 0, dpsd = 0)
pnmbf01(k = 10, n = 200, usd = 2, null = 0, psd = 0.5/sqrt(2), dpm = 0, dpsd = 0,
        lower.tail = FALSE)
```

powerbf01 *Power and sample size calculations for z-test Bayes factor*

Description

Compute probability that z-test Bayes factor is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

Usage

```
powerbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  sd = 1,
  null = 0,
  pm,
  psd,
  type = c("two.sample", "one.sample", "paired"),
  dpm = pm,
  dpsd = psd,
  nrange = c(1, 10^5),
  alternative = c("two.sided", "less", "greater"),
  tol = getOption("bfpwr.tol", 1e-08)
)
```

Arguments

n	Sample size (per group for two-sample tests). Has to be NULL if power is specified. Defaults to NULL
power	Target power. Has to be NULL if n is specified. Defaults to NULL
k	Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis
sd	Standard deviation of one observation (for type = "two.sample" or type = "one.sample") or of one difference within a pair of observations (type = "paired"). Is assumed to be known. Defaults to 1
null	Mean difference under the point null hypothesis. Defaults to 0
pm	Mean of the normal prior assigned to the mean difference under the alternative in the analysis
psd	Standard deviation of the normal prior assigned to the mean difference under the alternative in the analysis. Set to 0 to obtain a point prior at the prior mean
type	The type of test. One of "two.sample", "one.sample", "paired". Defaults to "two.sample"
dpm	Mean of the normal design prior assigned to the mean difference. Defaults to the same value as the analysis prior pm

dpsd	Standard deviation of the normal design prior assigned to the mean difference. Defaults to the same value as the analysis prior psd
nrange	Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(1, 10^5)
alternative	Direction of the alternative hypothesis, one of "two.sided" (default), "less", or "greater". For a one-sided alternative, the normal analysis prior is restricted to values below null ("less") or above null ("greater") and renormalized. pm and psd describe the normal distribution before truncation. A point prior (psd = 0) must lie strictly on the specified side of null.
tol	Numerical tolerance for sample-size root searches. Defaults to getOption("bfpwr.tol", 1e-8).

Details

This function provides a similar interface as stats::power.t.test. It also assumes that the data are continuous and that the parameter of interest is either a mean or a (standardized) mean difference. For some users, the low-level functions nbf01 (to directly compute the sample size for a fixed power) and pbf01 (to directly compute the power for a fixed sample size) may also be useful because they can be used for other data and parameter types. The sample-size tolerance tol is used only when a numerical search is required; an analytical solution does not use it.

Value

Object of class "power.bftest", a list of the arguments (including the computed one) augmented with method and note elements

Note

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors. The normal design prior remains untruncated for one-sided alternatives. Power need not increase with sample size; for example, the probability of misleading evidence first rises and then falls. If both endpoints of nrange are below the target, the numerical search checks the interior maximum. The returned sample size is the smallest at which power reaches the target; power may fall below the target again at larger sample sizes.

Author(s)

Samuel Pawel

See Also

plot.power.bftest, nbf01, pbf01, bf01

Examples

```
## determine power
powerbf01(n = 100, pm = 0, psd = 1, dpm = 0.5, dpsd = 0)

## determine sample size
```

```
powerbf01(power = 0.99, pm = 0, psd = 1, dpm = 0.5, dpsd = 0)

## one-sided alternative with a half-normal analysis prior
powerbf01(power = 0.8, pm = 0, psd = 1, dpm = 0.5, dpsd = 0,
          alternative = "greater")
```

`powerbf01seq` *Power and Maximum Sample Size Calculations for Sequential z-Test Bayes Factors*

Description

Computes cumulative stopping probabilities for a sequential z-test Bayes factor design, or determines the maximum sample size needed to obtain a target stopping probability.

Usage

```
powerbf01seq(
  n = NULL,
  power = NULL,
  k1 = 1/10,
  k0 = 1/k1,
  sd = 1,
  null = 0,
  pm,
  psd,
  type = c("two.sample", "one.sample", "paired"),
  bftype = c("normal", "directional", "moment"),
  dpm = pm,
  dpsd = psd,
  target = c("H1", "H0"),
  nrange = c(2, 10^5),
  looks = 1,
  timing = NULL,
  minN = NULL,
  by = NULL,
  strict = TRUE,
  search = c("adaptive", "exhaustive"),
  alternative = c("two.sided", "less", "greater"),
  ...
)
```

Arguments

`n`	Maximum sample size (per group for two-sample tests). Has to be `NULL` if `power` is specified. Defaults to `NULL`.

power — Target stopping probability.

k1 — Bayes factor threshold in favor of H_1. Evidence for H_1 is obtained when $\mathrm{BF}_{01} \leq k1$.

k0 — Bayes factor threshold in favor of H_0. Evidence for H_0 is obtained when $\mathrm{BF}_{01} \geq k0$.

sd — Standard deviation of one observation (for type = "two.sample" or type = "one.sample") or of one difference within a pair of observations (type = "paired"). Is assumed to be known. Defaults to 1.

null — Parameter value under the point null hypothesis. Defaults to 0

pm — Analysis prior mean. Not taken into account for bftype = "moment".

psd — Analysis prior standard deviation (bftype = "normal" and bftype = "directional") or scale (bftype = "moment").

type — Type of sampling design. One of "two.sample", "one.sample", or "paired". Defaults to "two.sample".

bftype — Type of z-test Bayes factor. One of "normal", "directional", or "moment". Defaults to "normal". One-sided alternative values support bftype = "normal" and bftype = "moment".

dpm — Mean of the normal design prior

dpsd — Standard deviation of the normal design prior. Set dpsd = 0 to obtain a point prior at dpm

target — Character string. Either "H1" for the final cumulative probability of stopping for H_1, or "H0" for the final cumulative probability of stopping for H_0.

nrange — Maximum sample size search range over which numerical search is performed. Defaults to c(2, 10^5).

looks — Number of sequential looks when timing is not supplied. The default looks = 1 gives a fixed design.

timing — Optional cumulative information fractions for the looks. Values must be positive, strictly increasing, no larger than one, and end at one. Candidate look sample sizes are ceiling(maximumN * timing - sqrt(.Machine$double.eps)) with the final look fixed to maximumN.

minN — First-look sample size when by is supplied. Defaults to the lower search bound.

by — Optional sample-size increase between looks. If supplied, schedules are generated as seq(minN, maximumN, by = by) with maximumN appended as the final look.

strict — Logical. If TRUE and there are more than two critical values per stage, enumerate all possible region combinations. If FALSE, only integrate over the main regions where the sign of the z-statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Region probabilities are numerically integrated under both settings. Defaults to TRUE

search — Sample-size search rule for schedules generated from looks/timing. "adaptive" uses bracketing and binary search with local checks. It is a heuristic that can skip

an isolated crossing on a non-monotone power curve, and stops with a diagnostic if a transient invalid candidate invalidates the bracket. Use `"exhaustive"` when the first crossing must be guaranteed. `"exhaustive"` scans the full feasible candidate maximum-sample-size range and returns the first finite candidate that reaches `power`, skipping transient invalid candidates and stopping at terminal invalid candidates. Ignored when `by` is supplied, where scheduled maximum sample sizes `minN + j * by` are scanned in increasing order. Defaults to `"adaptive"`.

`alternative` Direction of the alternative hypothesis, one of `"two.sided"` (default), `"less"`, or `"greater"`. For normal and moment priors, one-sided alternatives truncate and renormalize the analysis prior below or above `null`, respectively, while retaining the point null hypothesis. Prior parameters describe the distribution before truncation. A normal point prior (`psd = 0`) must lie strictly on the specified side of `null`.

`...` Additional arguments passed to `pbf01seq`.

Details

This function provides a higher-level interface to `pbf01seq` and `nbf01seq` for continuous data. If `power` is supplied, the returned design is evaluated at the searched maximum sample size, and `solver$reached` records whether the target was achieved within `nrange`. If `n` is supplied, no search is performed and the `solver` element records the achieved stopping probability for the fixed schedule. For fixed-`n` increment schedules with missing `minN`, `nrange[1]` is used as the default first look, clamped to `n` when necessary.

Value

An object of class `"bfseqdesign"` containing the sequential design, augmented with a `solver` element. In fixed-`n` mode, `solver$targetPower` and `solver$reached` are `NA`.

Author(s)

František Bartoš

See Also

pbf01seq, nbf01seq, powerbf01

Examples

```
powerbf01seq(n = 60, k1 = 1/5, k0 = 5, pm = 0, psd = 1,
             dpm = 0.5, dpsd = 0, looks = 3)
powerbf01seq(power = 0.8, k1 = 1/5, k0 = 5, pm = 0, psd = 1,
             dpm = 0.5, dpsd = 0, looks = 3, nrange = c(2, 200))

## point null versus a positive half-normal alternative
powerbf01seq(power = 0.8, pm = 0, psd = 1, dpm = 0.5, dpsd = 0,
             alternative = "greater", looks = 3, nrange = c(2, 200))

## point null versus a positive normal moment alternative
```

```
powerbf01seq(n = 100, psd = 0.5/sqrt(2), dpm = 0.5, dpsd = 0,
             bftype = "moment", alternative = "greater", looks = 3)
```

---

powerbinbf01 *Power and sample size calculations for binomial Bayes factor*

Description

Compute probability that binomial Bayes factor (binbf01) is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

Usage

```
powerbinbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  p0 = 0.5,
  type = c("point", "direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1,
  nrange = c(1, 10^4),
  tol = getOption("bfpwr.tol", 1e-08)
)
```

Arguments

n	Sample size. Has to be NULL if power is specified. Defaults to NULL
power	Target power. Has to be NULL if n is specified. Defaults to NULL
k	Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis
p0	Tested binomial proportion. Defaults to 0.5
type	Type of test. Can be "point" or "direction". Defaults to "point"
a	Number of successes parameter of the beta analysis prior distribution. Defaults to 1
b	Number of failures parameter of the beta analysis prior distribution. Defaults to 1
dp	Fixed binomial proportion assumed for the power calculation. Set to NA to use a truncated beta design prior instead (specified via the da, db, dl, and du arguments). Defaults to NA

da	Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior
db	Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior
dl	Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0
du	Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1
nrange	Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(1, 10^3)
tol	Numerical tolerance for sample-size root searches. Defaults to getOption("bfpwr.tol", 1e-8).

Details

This function provides a similar interface as stats::power.prop.test. For some users, the low-level functions nbinbf01 (to directly compute the sample size for a fixed power) and pbinbf01 (to directly compute the power for a fixed sample size) may also be useful.

Value

Object of class "power.bftest", a list of the arguments (including the computed one) augmented with method and note elements

Author(s)

Samuel Pawel

See Also

plot.power.bftest, pbinbf01, nbinbf01, binbf01

Examples

```
## determine sample size
(nres <- powerbinbf01(power = 0.8, p0 = 0.2, type = "direction", dl = 0.2))
## Not run:
plot(nres, nlim = c(1, 250), ngrid = 250, type = "s")

## End(Not run)

## determine power
(powres <- powerbinbf01(n = 100, type = "point"))
## Not run:
plot(powres)

## End(Not run)
```

`powerbinbf01seq` *Power calculations for binomial Bayes factor with two-stage designs*

Description

Compute design characteristics for two-stage binomial Bayes factor design which stops for futility if the binomial Bayes factor (binbf01) is equal or below a threshold at final stage, taking into account that the study can stop if it is greater than a specified threshold at interim.

Usage

```
powerbinbf01seq(
  n1,
  n2,
  k = 1/10,
  kf = 1/k,
  p0 = 0.5,
  type = c("direction"),
  a = 1,
  b = 1,
  dp = NA,
  da = a,
  db = b,
  dl = 0,
  du = 1
)
```

Arguments

`n1`	Sample size at interim stage
`n2`	Sample size at final stage
`k`	Bayes factor threshold for evidence for efficacy. Defaults to `1/10`, Jeffreys' threshold for 'strong evidence' against the null hypothesis
`kf`	Bayes factor threshold for evidence for futility. Defaults to `1/k`.
`p0`	Tested binomial proportion. Defaults to `0.5`
`type`	Type of test. Currently only `"direction"` is implemented
`a`	Number of successes parameter of the beta analysis prior distribution. Defaults to `1`
`b`	Number of failures parameter of the beta analysis prior distribution. Defaults to `1`
`dp`	Fixed binomial proportion assumed for the power calculation. Set to `NA` to use a truncated beta design prior instead (specified via the `da`, `db`, `dl`, and `du` arguments). Defaults to `NA`

da	Number of successes parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value a as specified for the analysis prior
db	Number of failures parameter of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to the same value b as specified for the analysis prior
dl	Lower truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 0
du	Upper truncation limit of of the truncated beta design prior distribution. Is only taken into account if dp = NA. Defaults to 1

Details

For some users, the low-level function pbinbf01seq that is used internally may also be useful.

Value

An object of type "power.bftestseq", i.e., a list of the arguments, computed probabilities and expected samples sizes, and method and note elements

Author(s)

Samuel Pawel

See Also

binbf01, pbinbf01seq

Examples

```
## point prior under the alternative
powerbinbf01seq(n1 = 20, n2 = 40, k = 1/10, kf = 3, p0 = 0.2, dp = 0.4)

## uniform prior [0.2, 1] under the alternative
powerbinbf01seq(n1 = 20, n2 = 40, k = 1/10, kf = 3, p0 = 0.2, dl = 0.2, du = 1)
```

---

powernmbf01 *Power and sample size calculations for normal moment prior Bayes factor*

Description

Compute probability that normal moment prior Bayes factor is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

Usage

```
powernmbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  sd = 1,
  null = 0,
  psd,
  type = c("two.sample", "one.sample", "paired"),
  dpm,
  dpsd,
  nrange = c(1, 10^5),
  tol = getOption("bfpwr.tol", 1e-08),
  alternative = c("two.sided", "less", "greater")
)
```

Arguments

n	Sample size (per group for two-sample tests). Has to be NULL if power is specified. Defaults to NULL
power	Target power. Has to be NULL if n is specified. Defaults to NULL
k	Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis
sd	Standard deviation of one observation (for type = "two.sample" or type = "one.sample") or of one difference within a pair of observations (type = "paired"). Is assumed to be known. Defaults to 1
null	Parameter value under the point null hypothesis. Defaults to 0
psd	Spread of the normal moment prior assigned to the parameter under the alternative in the analysis. The modes of the prior are located at $\pm\sqrt{2}$ psd
type	The type of test. One of "two.sample", "one.sample", "paired". Defaults to "two.sample"
dpm	Mean of the normal design prior assigned to the parameter
dpsd	Standard deviation of the normal design prior assigned to the parameter. Set to 0 to obtain a point prior at the design prior mean
nrange	Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(1, 10^5)
tol	Numerical tolerance for sample-size root searches. Defaults to getOption("bfpwr.tol", 1e-8).
alternative	Direction of the alternative hypothesis, one of "two.sided" (default), "less", or "greater". For a one-sided alternative, the normal moment analysis prior is restricted to values below null ("less") or above null ("greater") and renormalized. The point null hypothesis is unchanged; psd is the scale before truncation.

Details

This function provides a similar interface as `stats::power.t.test`. It also assumes that the data are continuous and that the parameter of interest is either a mean or a (standardized) mean difference. For some users, the low-level functions nnmbf01 (to directly compute the sample size for a fixed power) and pnmbf01 (to directly compute the power for a fixed sample size) may also be useful because they can be used for other data and parameter types.

Value

Object of class `"power.bftest"`, a list of the arguments (including the computed one) augmented with `method` and `note` elements

Note

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors. The normal design prior remains untruncated for one-sided alternatives. Power need not increase with sample size; for example, the probability of misleading evidence first rises and then falls. If both endpoints of `nrange` are below the target, the numerical search checks the interior maximum. The returned sample size is the smallest at which power reaches the target; power may fall below the target again at larger sample sizes.

Author(s)

Samuel Pawel

See Also

plot.power.bftest, nnmbf01, pnmbf01, nmbf01

Examples

```
## determine power
powernmbf01(n = 100, psd = 1, dpm = 0.5, dpsd = 0)

## determine sample size
powernmbf01(power = 0.99, psd = 1, dpm = 0.5, dpsd = 0)
```

`powertbf01` *Power and sample size calculations for t-test Bayes factor*

Description

Compute probability that t-test Bayes factor is smaller than a specified threshold (the power), or determine sample size to obtain a target power.

Usage

```
powertbf01(
  n = NULL,
  power = NULL,
  k = 1/10,
  null = 0,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  dpm = plocation,
  dpsd = pscale,
  nrange = c(2, 10^4),
  ratio = 1,
  drange = "adaptive",
  tail.eps = getOption("bfpwr.tail.eps", 1e-06),
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  ...
)
```

Arguments

n	Sample size. For two-sample designs with unequal allocation, n is the group-1 sample size and group 2 is evaluated as ceiling(n * ratio). Has to be NULL if power is specified. Defaults to NULL
power	Target power. Has to be NULL if n is specified. Defaults to NULL
k	Bayes factor threshold. Defaults to 1/10, Jeffreys' threshold for 'strong evidence' against the null hypothesis
null	Standardized mean difference under the point null hypothesis. Defaults to 0
plocation	t prior location. Defaults to 0
pscale	t prior scale. Defaults to 1/sqrt(2)
pdf	t prior degrees of freedom. Defaults to 1 (a Cauchy prior)
type	Type of t-test. Can be "two.sample" (default), "one.sample", or "paired"
alternative	Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to effects below and above null, respectively.
dpm	Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location
dpsd	Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale
nrange	Sample size search range over which numerical search is performed (only taken into account when n is NULL). Defaults to c(2, 10^4)

`ratio` Allocation ratio `n2 / n1` for two-sample designs. Candidate group-2 sample sizes are `ceiling(n1 * ratio)`. Ignored for one-sample and paired designs.

`drange` Standardized mean difference search range over which the critical values are searched for. Can be either set to a numerical range or to `"adaptive"` (default) which determines the range in an adaptive way from the other input parameters

`tail.eps` One-sided adaptive searches stop once the remaining predictive probability in the searched tail is at most this value. If no boundary is found before that point, the boundary-free probability implied by the searched side is returned and the omitted tail probability is bounded by `tail.eps`. Smaller values search farther and can recover extremely remote boundaries at additional computational cost. Defaults to `1e-6`

`tail.nquad` Number of Gauss-Legendre quadrature nodes used by `tbf01` for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to `512`.

`...` Numerical controls inherited from `ptbf01`, also used by `ntbf01` when finding a sample size. Saved in the returned object and reused when plotting power curves.

Details

This function provides a similar interface as `stats::power.t.test`. For some users, the low-level functions ntbf01 (to directly compute the sample size for a fixed power) and ptbf01 (to directly compute the power for a fixed sample size) may also be useful.

Value

Object of class `"power.bftest"`, a list of the arguments (including the computed one) augmented with `method` and `note` elements

Note

A warning message will be displayed in case that the specified target power is not achievable under the specified analysis and design priors. The normal design prior remains untruncated for one-sided alternatives. Power need not increase with sample size; for example, the probability of misleading evidence first rises and then falls. If both endpoints of `nrange` are below the target, the numerical search checks the interior maximum. The returned sample size is the smallest at which power reaches the target; power may fall below the target again at larger sample sizes.

Author(s)

Samuel Pawel, František Bartoš

See Also

plot.power.bftest, ptbf01, ntbf01, tbf01

Examples

```
## determine power
powertbf01(n = 146, k = 1/6, dpm = 0.5, dps = 0, alternative = "greater")

## determine sample size
powertbf01(power = 0.95, k = 1/6, dpm = 0.5, dps = 0, alternative = "greater")
```

`powertbf01seq` *Power and Maximum Sample Size Calculations for Sequential t-Test Bayes Factors*

Description

Computes cumulative stopping probabilities for a sequential t-test Bayes factor design, or determines the maximum sample size needed to obtain a target stopping probability.

Usage

```
powertbf01seq(
  n = NULL,
  power = NULL,
  k1 = 1/10,
  k0 = 1/k1,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  dpm = plocation,
  dpsd = pscale,
  target = c("H1", "H0"),
  nrange = c(2, 10^4),
  looks = 1,
  timing = NULL,
  minN = NULL,
  by = NULL,
  ratio = 1,
  strict = TRUE,
  trange = "adaptive",
  tail.eps = getOption("bfpwr.tail.eps", 1e-06),
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  search = c("adaptive", "exhaustive"),
  ...
)
```

Arguments

n	Maximum sample size in group 1 for two-sample tests, or maximum sample size for one-sample and paired tests. Has to be NULL if power is specified. Defaults to NULL.
power	Target stopping probability.
k1	Bayes factor threshold in favor of H_1. Evidence for H_1 is obtained when $\text{BF}_{01} \leq k1$.
k0	Bayes factor threshold in favor of H_0. Evidence for H_0 is obtained when $\text{BF}_{01} \geq k0$.
plocation	t prior location. Defaults to 0
pscale	t prior scale. Defaults to 1/sqrt(2)
pdf	t prior degrees of freedom. Defaults to 1 (a Cauchy prior)
type	Type of t-test. Can be "two.sample" (default), "one.sample", or "paired"
alternative	Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to effects below and above null, respectively.
dpm	Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location
dpsd	Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale
target	Character string. Either "H1" for the final cumulative probability of stopping for H_1, or "H0" for the final cumulative probability of stopping for H_0.
nrange	Maximum sample size search range over which numerical search is performed. Defaults to c(2, 10^4).
looks	Number of sequential looks when timing is not supplied. The default looks = 1 gives a fixed design.
timing	Optional cumulative information fractions for the looks. Values must be positive, strictly increasing, no larger than one, and end at one. Candidate look sample sizes are ceiling(maximumN * timing - sqrt(.Machine$double.eps)) with the final look fixed to maximumN.
minN	First-look sample size when by is supplied. Defaults to the lower search bound.
by	Optional sample-size increase between looks. If supplied, schedules are generated as seq(minN, maximumN, by = by) with maximumN appended as the final look.
ratio	Allocation ratio n2 / n1 for two-sample designs. Candidate group-2 sample sizes are ceiling(n1 * ratio). Ignored for one-sample and paired designs.
strict	Logical. If TRUE and there are more than two critical values per stage, enumerate all possible region combinations. If FALSE, only integrate over the main regions where the sign of the z-statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Region probabilities are numerically integrated under both settings. Defaults to TRUE

`trange`	Critical t-statistic search strategy for the sequential stopping boundaries. Can be either `"adaptive"` (default) or a numeric interval. For one-sided adaptive searches, `tail.eps` determines the predictive tail cutoff used to stop finite scans. Use a numeric interval to force exact boundary searches over that interval.
`tail.eps`	One-sided adaptive sequential boundary searches stop once the marginal predictive tail probability beyond the searched t statistic is at most `tail.eps`. The same cutoff is used for each adaptive boundary search; it is not split across looks or boundaries. If no boundary is found before the cutoff, it is placed at infinity in the searched direction, preserving the H0 and H1 sides of the stopping event. The omitted marginal mass is bounded by `tail.eps`; with C unresolved boundary searches, a conservative combined bound is `min(1, C * tail.eps)` rather than `tail.eps`. smaller values search farther at additional computational cost. Defaults to `1e-6`.
`tail.nquad`	Number of Gauss-Legendre quadrature nodes used by `tbf01` for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to `512`.
`search`	Sample-size search rule for schedules generated from `looks`/`timing`. `"adaptive"` uses bracketing and binary search with local checks. It is a heuristic that can skip an isolated crossing on a non-monotone power curve, and stops with a diagnostic if a transient invalid candidate invalidates the bracket. Use `"exhaustive"` when the first crossing must be guaranteed. `"exhaustive"` scans the full feasible candidate maximum-sample-size range and returns the first finite candidate that reaches `power`, skipping transient invalid candidates and stopping at terminal invalid candidates. Ignored when `by` is supplied, where scheduled maximum sample sizes `minN + j * by` are scanned in increasing order. Defaults to `"adaptive"`.
`...`	Additional arguments passed to `ptbf01seq`.

Details

This function provides a higher-level interface to `ptbf01seq` and `ntbf01seq`. If `power` is supplied, the returned design is evaluated at the searched maximum sample size, and `solver$reached` records whether the target was achieved within `nrange`. If `n` is supplied, no search is performed and the `solver` element records the achieved stopping probability for the fixed schedule. For fixed-`n` increment schedules with missing `minN`, `nrange[1]` is used as the default first look, clamped to `n` when necessary.

Value

An object of class `"bfseqdesign"` containing the sequential design, augmented with a `solver` element. In fixed-`n` mode, `solver$targetPower` and `solver$reached` are `NA`.

Author(s)

František Bartoš

See Also

ptbf01seq, ntbf01seq, powertbf01

Examples

```
powertbf01seq(n = 20, k1 = 1/2, k0 = 2, dpm = 0.5, dpsd = 0,
              alternative = "greater", looks = 2, strict = FALSE)
powertbf01seq(power = 0.4, k1 = 1/2, k0 = 2, dpm = 0.5, dpsd = 0,
              alternative = "greater", looks = 2, nrange = c(2, 80),
              strict = FALSE)
```

---

print.bfseqdesign *Print method for class* "bfseqdesign"

Description

Print method for class "bfseqdesign"

Usage

```
## S3 method for class 'bfseqdesign'
print(x, digits = max(3L, getOption("digits") - 3L), ...)
```

Arguments

x	Object of class "bfseqdesign"
digits	Number of digits for formatting of numbers
...	Other arguments (for consistency with the generic)

Value

Prints text summary in the console and invisibly returns the "bfseqdesign" object

Author(s)

Samuel Pawel, František Bartoš

See Also

pbf01seq

Examples

```
n <- seq(50, 300, 50) # sample size per stage
se <- sqrt(2/n) # standard errors per stage
res <- pbf01seq(k1 = 1/10, k0 = 5, se = se, n = n, pm = 0, psd = 1,
                dpm = 0.5, dpsd = 0.1, type = "normal")
res
```

`print.power.bftest` *Print method for class* `"power.bftest"`

Description

Print method for class `"power.bftest"`

Usage

```
## S3 method for class 'power.bftest'
print(x, digits = getOption("digits"), ...)
```

Arguments

`x`	Object of class `"power.bftest"`
`digits`	Number of digits for formatting of numbers
`...`	Other arguments (for consistency with the generic)

Value

Prints text summary in the console and invisibly returns the `"power.bftest"` object

Note

Function adapted from `stats:::print.power.htest` written by Peter Dalgaard

Author(s)

Samuel Pawel, František Bartoš

See Also

powerbf01

Examples

```
powerbf01(power = 0.95, pm = 0, psd = 1, dpm = 0.5, dpsd = 0)
powerbf01(power = 0.95, pm = 0, psd = 1, dpm = 0.5, dpsd = 0, type = "one.sample")
powerbf01(power = 0.95, pm = 0, psd = 1, dpm = 0.5, dpsd = 0, type = "paired")
powerbf01(power = 0.95, pm = 1, psd = 0, dpm = 0.8, dpsd = 0, type = "paired")
```

print.power.bftestseq *Print method for class* "power.bftestseq"

Description

Print method for class "power.bftestseq"

Usage

```
## S3 method for class 'power.bftestseq'
print(x, digits = getOption("digits"), ...)
```

Arguments

x	Object of class "power.bftestseq"
digits	Number of digits for formatting of numbers
...	Other arguments (for consistency with the generic)

Value

Prints text summary in the console and invisibly returns the "power.bftestseq" object

Author(s)

Samuel Pawel

ptbf01 *Cumulative distribution function of the t-test Bayes factor*

Description

This function computes the probability of obtaining a t-test Bayes factor (tbf01) more extreme than a threshold k with a specified sample size. The future standardized estimate is modeled with its known-variance normal distribution. Thus, although the analysis Bayes factor uses the finite-sample t distribution, the returned power is an asymptotic approximation rather than exact noncentral-t power; differences can be material at small sample sizes.

Usage

```
ptbf01(
  k,
  n,
  n1 = n,
  n2 = n,
  null = 0,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  dpm = plocation,
  dpsd = pscale,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  lower.tail = TRUE,
  drange = "adaptive",
  tail.eps = getOption("bfpwr.tail.eps", 1e-06),
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  ...
)
```

Arguments

k	Bayes factor threshold
n	Sample size (per group)
n1	Sample size in group 1 (only required for two-sample t-test with unequal group sizes)
n2	Sample size in group 2 (only required for two-sample t-test with unequal group sizes)
null	Standardized mean difference under the point null hypothesis. Defaults to 0
plocation	t prior location. Defaults to 0
pscale	t prior scale. Defaults to 1/sqrt(2)
pdf	t prior degrees of freedom. Defaults to 1 (a Cauchy prior)
dpm	Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location
dpsd	Standard deviation of the normal design prior assigned to the standardized mean difference. Set to 0 to obtain a point prior at the design prior mean. Defaults to the analysis prior scale
type	Type of t-test. Can be "two.sample" (default), "one.sample", or "paired"
alternative	Direction of the test. Can be either "two.sided" (default), "less", or "greater". The latter two truncate the analysis prior to effects below and above null, respectively.
lower.tail	Logical indicating whether Pr($\mathrm{BF}_{01} \leq$ k) (TRUE) or Pr($\mathrm{BF}_{01} >$ k) (FALSE) should be computed. Defaults to TRUE

drange	Standardized mean difference search range over which the critical values are searched for. Can be either set to a numerical range or to "adaptive" (default) which determines the range in an adaptive way from the other input parameters
tail.eps	One-sided adaptive searches stop once the remaining predictive probability in the searched tail is at most this value. If no boundary is found before that point, the boundary-free probability implied by the searched side is returned and the omitted tail probability is bounded by tail.eps. Smaller values search farther and can recover extremely remote boundaries at additional computational cost. Defaults to 1e-6
tail.nquad	Number of Gauss-Legendre quadrature nodes used by tbf01 for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to 512.
...	Optional numerical controls for all boundary searches. rel.tol (default 1e-8), abs.tol (default rel.tol), and subdivisions (default 1000), together with stop.on.error and keep.xy, are passed to stats::integrate for BF evaluation. Root searches use tol (default 1e-8), maxiter, trace, and check.conv from stats::uniroot. One-sided scouting uses a looser integral unless rel.tol is supplied; returned roots are checked against the final integral with the requested accuracy. Session defaults can be changed with bfpwrOptions. These controls reduce numerical error within the normal predictive approximation; they do not remove that approximation.

Value

The probability that the Bayes factor is less or greater (depending on the specified lower.tail) than the specified threshold k

Author(s)

Samuel Pawel, František Bartoš

See Also

tbf01, ntbf01, powertbf01

Examples

```
## example from Schönbrodt and Wagenmakers (2018, p. 135)
ptbf01(k = 1/6, n = 146, dpm = 0.5, dpsd = 0, alternative = "greater")
ptbf01(k = 6, n = 146, dpm = 0, dpsd = 0, alternative = "greater",
       lower.tail = FALSE)

## two-sided
ptbf01(k = 1/6, n = 146, dpm = 0.5, dpsd = 0)
ptbf01(k = 6, n = 146, dpm = 0, dpsd = 0, lower.tail = FALSE)

## one-sample test
ptbf01(k = 1/6, n = 146, dpm = 0.5, dpsd = 0, alternative = "greater",
       type = "one.sample")
```

ptbf01seq *Sequential T-Test Bayes Factor Design*

Description

Computes cumulative probabilities of observing t-test Bayes factors that provide evidence for the null hypothesis H_0, the alternative hypothesis H_1, or remain inconclusive in a sequential design. Also computes the expected sample size.

Usage

```
ptbf01seq(
  k1,
  k0 = 1/k1,
  n,
  n1 = n,
  n2 = n,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  dpm = plocation,
  dpsd = pscale,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  strict = TRUE,
  trange = "adaptive",
  tail.eps = getOption("bfpwr.tail.eps", 1e-06),
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  ...
)
```

Arguments

k1	Bayes factor threshold in favor of H_1 (i.e., $\text{BF}_{01} \leq \texttt{k1} < 1$ implies evidence for H_1)
k0	Bayes factor threshold in favor of H_0 (i.e., $\text{BF}_{01} \geq \texttt{k0} > 1$ implies evidence for H_0)
n	Sample size (per group)
n1	Sample size in group 1 (only required for two-sample t-test with unequal group sizes)
n2	Sample size in group 2 (only required for two-sample t-test with unequal group sizes)
plocation	t prior location. Defaults to `0`
pscale	t prior scale. Defaults to `1/sqrt(2)`
pdf	t prior degrees of freedom. Defaults to `1` (a Cauchy prior)

`dpm` Mean of the normal design prior assigned to the standardized mean difference. Defaults to the analysis prior location

`dpsd` Standard deviation of the normal design prior assigned to the standardized mean difference. Set to `0` to obtain a point prior at the design prior mean. Defaults to the analysis prior scale

`type` Type of t-test. Can be `"two.sample"` (default), `"one.sample"`, or `"paired"`

`alternative` Direction of the test. Can be either `"two.sided"` (default), `"less"`, or `"greater"`. The latter two truncate the analysis prior to effects below and above `null`, respectively.

`strict` Logical. If `TRUE` and there are more than two critical values per stage, enumerate all possible region combinations. If `FALSE`, only integrate over the main regions where the sign of the z-statistics does not change across stages (faster, recommended when many interim analyses, e.g., more than 10, are performed). Region probabilities are numerically integrated under both settings. Defaults to `TRUE`

`trange` Critical t-statistic search strategy for the sequential stopping boundaries. Can be either `"adaptive"` (default) or a numeric interval. For one-sided adaptive searches, `tail.eps` determines the predictive tail cutoff used to stop finite scans. Use a numeric interval to force exact boundary searches over that interval.

`tail.eps` One-sided adaptive sequential boundary searches stop once the marginal predictive tail probability beyond the searched t statistic is at most `tail.eps`. The same cutoff is used for each adaptive boundary search; it is not split across looks or boundaries. If no boundary is found before the cutoff, it is placed at infinity in the searched direction, preserving the H0 and H1 sides of the stopping event. The omitted marginal mass is bounded by `tail.eps`; with C unresolved boundary searches, a conservative combined bound is `min(1, C * tail.eps)` rather than `tail.eps`. smaller values search farther at additional computational cost. Defaults to `1e-6`.

`tail.nquad` Number of Gauss-Legendre quadrature nodes used by `tbf01` for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to `512`.

`...` Numerical integration controls inherited from `pbf01seq`, including `ngrid`. Pass BF integration and critical-value root controls from `ptbf01` in a named list, for example `bf.control = list(rel.tol = 1e-9, tol = 1e-9)`. This keeps the root tolerance separate from `mvtnorm`'s `tol`.

Details

The function constructs per-stage integration regions for cumulative z-statistics based on the Bayes factor thresholds `k1` and `k0`, then computes the probability of these regions under a predictive distribution defined by the known-variance, asymptotic approximation to the variance of the t-statistic and the normal design prior with `dpm` and `dpsd`. Integration is performed via `mvtnorm::lpmvnorm`. This is deterministic numerical integration; increasing `ngrid` can be used to check convergence.

Value

An object of class "bfseqdesign", which is a list containing the input arguments, the critical z-values, the expected sample size, the cumulative probabilities of stopping for H_1 and H_0 by each stage, and the cumulative probabilities of remaining inconclusive by each stage.

Author(s)

Samuel Pawel, František Bartoš

Examples

```
## similar to example from Schönbrodt and Wagenmakers (2018, p. 138)
k0 <- 6
k1 <- 1/30
dpm <- 0.5
dpsd <- 0.1
plocation <- 0
pscale <- 1/sqrt(2)
pdf <- 1
type <- "two.sample"
alternative <- "greater"
n <- seq(40, 100, 10) # sample size (per group) per stage
res <- ptbf01seq(k1 = k1, k0 = k0, n = n, plocation = plocation,
                 pscale = pscale, pdf = pdf, dpm = dpm, dpsd = dpsd,
                 alternative = alternative, type = type)
res
plot(res) # show stopping probabilities
plot(res, zplot = TRUE) # show critical z-values
```

---

tbf01 *t-test Bayes factor*

Description

This function computes the Bayes factor that forms the basis of the informed Bayesian t-test from Gronau et al. (2020). The Bayes factor quantifies the evidence that the data provide for the null hypothesis that the standardized mean difference (SMD) is zero against the alternative that the SMD is non-zero. A location-scale t-distribution is assumed for the SMD under the alternative hypothesis. The Jeffreys-Zellner-Siow (JZS) Bayes factor (Rouder et al., 2009) is obtained as a special case by setting the location of the prior to zero and the prior degrees of freedom to one, which is the default.

The data are summarized by t-statistics and sample sizes. The following types of t-statistics are accepted:

- Two-sample t-test where the SMD represents the standardized mean difference between two group means (assuming equal variances in both groups)
- One-sample t-test where the SMD represents the standardized mean difference to the null value
- Paired t-test where the SMD represents the standardized mean change score

Usage

```
tbf01(
  t,
  n,
  n1 = n,
  n2 = n,
  plocation = 0,
  pscale = 1/sqrt(2),
  pdf = 1,
  type = c("two.sample", "one.sample", "paired"),
  alternative = c("two.sided", "less", "greater"),
  log = FALSE,
  tail.nquad = getOption("bfpwr.tail.nquad", 512),
  ...
)
```

Arguments

`t`	t-statistic
`n`	Sample size (per group)
`n1`	Sample size in group 1 (only required for two-sample t-test with unequal group sizes)
`n2`	Sample size in group 2 (only required for two-sample t-test with unequal group sizes)
`plocation`	t prior location. Defaults to `0`
`pscale`	t prior scale. Defaults to `1/sqrt(2)`
`pdf`	t prior degrees of freedom. Defaults to `1` (a Cauchy prior)
`type`	Type of t-test. Can be `"two.sample"` (default), `"one.sample"`, or `"paired"`
`alternative`	Direction of the test. Can be either `"two.sided"` (default), `"less"`, or `"greater"`. The latter two truncate the analysis prior to negative and positive effects, respectively.
`log`	Logical indicating whether the natural logarithm of the Bayes factor should be returned. Defaults to `FALSE`
`tail.nquad`	Number of Gauss-Legendre quadrature nodes used for stable wrong-tail one-sided calculations. Larger values are more accurate but slower. Defaults to `512` nodes per dimension (the fallback uses a two-dimensional product rule).
`...`	Additional arguments passed to `stats::integrate` for the direct one-dimensional Bayes factor integral. Defaults are `rel.tol = 1e-8`, `abs.tol = rel.tol`, and `subdivisions = 1000`. Session defaults can be changed with bfpwrOptions.

Details

The Bayes factor is implemented as in equation (5) in Gronau et al. (2020), and using suitable truncation in case of one-sided alternatives. Most calculations use `stats::integrate`. Extreme wrong-tail one-sided calculations use fixed Gauss-Legendre quadrature to avoid noncentral-t underflow.

Value

Bayes factor in favor of the null hypothesis over the alternative ($BF_{01} > 1$ indicates evidence for the null hypothesis, whereas $BF_{01} < 1$ indicates evidence for the alternative)

Author(s)

Samuel Pawel, František Bartoš

References

Rouder, J. N., Speckman, P. L., Sun, D., Morey, R. D., Iverson, G. (2009). Bayesian t tests for accepting and rejecting the null hypothesis. Psychonomic Bulletin & Review, 16(2):225-237. doi:10.3758/PBR.16.2.225

Gronau, Q. F., Ly., A., Wagenmakers, E.J. (2020). Informed Bayesian t-Tests. The American Statistician, 74(2):137-143. doi:10.1080/00031305.2018.1562983

See Also

powertbf01, ptbf01, ntbf01

Examples

```
## analyses from Rouder et al. (2009):
## values from Table 1
tbf01(t = c(0.69, 3.20), n = 100, pscale = 1, type = "one.sample")
## examples from p. 232
tbf01(t = c(2.24, 2.03), n = 80, pscale = 1, type = "one.sample")

## analyses from Gronau et al. (2020) section 3.2:
## informed prior
tbf01(t = -0.90, n1 = 53, n2 = 57, plocation = 0.350, pscale = 0.102, pdf = 3,
      alternative = "greater", type = "two.sample")
## default (one-sided) prior
tbf01(t = -0.90, n1 = 53, n2 = 57, plocation = 0, pscale = 1/sqrt(2), pdf = 1,
      alternative = "greater", type = "two.sample")
```

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